

破轮 is a uniform disk, so its moment of inertia is $I = \frac{1}{2}MR^2$. When rolling without slipping on the ground, the total kinetic energy is the sum of translational and rotational parts:

$$E_k = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv^2 + \frac{1}{2} \cdot \frac{1}{2}MR^2\left(\frac{v}{R}\right)^2 = \frac{3}{4}Mv^2.$$

Q1. Kinetic energy lost. As 破轮 rolls from the ground (centre of mass at height R) up to the top of the hemisphere, its centre of mass rises to $R + r$, an increase of $\Delta h = r$. For ideal rolling without slipping there is no dissipation, so the kinetic energy lost is entirely converted into gravitational potential energy:

$$\boxed{\Delta E_k = Mgr}.$$

Q2. Minimum initial speed. To *just barely* roll over Professor 孙笑川, 破轮 arrives at the top of the hemisphere with zero kinetic energy. By conservation of mechanical energy,

$$\frac{3}{4}Mv_{\min}^2 = Mgr \implies \boxed{v_{\min} = \sqrt{\frac{4}{3}gr}}.$$